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The Wonderful World of Science

NCF features

- Class Discussions
- Higher Order Thinking
- Activity-based Learning



THINK AND DISCUSS! Reflective thinking

- ◆ Are you curious about the world around you?
- ◆ Do you like to learn about how things work?
- ◆ Do you try to come up with your own solution to everyday problems?

If you answered yes to these questions, then you might already think like a scientist! The most important thing you should possess to think like a scientist is **curiosity** about the things around you.

WHAT IS SCIENCE?

Humans have always been curious to know more about the living and non-living things around them, and the various natural phenomena they can observe such as the formation of day and night, change of weather, and occurrence of seasons. The constant efforts to understand the working of nature, through centuries of careful observation and analysis, has given birth to **science** and its various branches.

Science is everywhere—from the deepest oceans to the most distant stars and planets,

from the tiniest living thing to the mammoth extinct dinosaurs, from the cooking of food to the flying of supersonic jets.

Science is the process of acquiring knowledge about the world we live in by asking questions, observing, doing experiments and reasoning out to find answers. This process has been going on for centuries and will continue for ever, because knowledge can never be complete. The more we discover, the more questions we find, the answers to which are not known.

How science develops

Let us illustrate this with an example. Long, long ago people had no idea what the sun was and how it got its heat. The nature of the sun was a **problem** for them. But since they realized that their existence was dependent on the sun, they worshipped it as god. But some people had more curious minds. They were not satisfied with the explanation that the sun was a god. Little by little as they made **observations** and collected **information** about the sun and other heavenly bodies, they realized that the sun was just one of

IN THIS CHAPTER

THE MEANING OF 'SCIENCE' AND 'SCIENTIFIC METHOD' ◆ USES OF SCIENTIFIC METHOD TO SOLVE DIFFERENT PROBLEMS—SCIENTIFIC OR OTHERWISE ◆ USE OF SCIENCE AS A TOOL TO BENEFIT HUMAN BEINGS

the billions and billions of stars in the universe. It just happened to be closer to us than the other stars. So it wasn't so unique after all!

Next came the question of how the sun got its heat. Some people said that there was a huge amount of coal on the sun which was burning. This was an assumption or **hypothesis** which offered a possible explanation to the problem. But it didn't take long to disprove this hypothesis. It was calculated that if this was so, the sun would burn out in 5000 years. And we know that it has been there for a much longer time and shows no sign of burning out.

Further observations were made to collect more information on the nature of light emitted by the sun. From this, scientists discovered that the sun consists mostly of the hydrogen gas. This baffled them. How could so much energy come from hydrogen?

In the meantime, some other scientists were carrying out experiments with particles that existed inside atoms. They discovered that under certain circumstances, atoms could give out huge amounts of energy. This happened when the particles present inside the atoms were disturbed in certain ways. Could this also be the source of energy of the sun?

Scientists then had another hypothesis that the hydrogen atoms in the sun were constantly coming together to form a gas called helium. During this process, changes took place inside the atoms and huge amount of energy was given out. This was the source of the sun's energy. Further observations and **experiments** proved that this hypothesis was

correct. It then became the conclusion or the **theory**.

Having learnt so much about what was happening in the sun, scientists next wanted to **apply** this knowledge and see if they could carry out similar processes on the earth. They succeeded and made the atomic and the hydrogen bomb. In the hydrogen bomb, which is more powerful than the atomic bomb, the processes going on in the sun were duplicated.

Did the scientists stop after that? Of course they didn't. They now wanted to control this energy so that it could be used to do some useful work, like generating electricity. They have succeeded in using the principle of the atomic bomb to generate electricity. This is an important and useful **application** of the theory discussed earlier. But till now it has not been possible to control the energy in the hydrogen bomb.

Scientists all over the world are trying to control the large amount of energy given out when hydrogen atoms combine to form helium atoms. Some day this may become possible, and when it does, it may solve all our energy problems!

THE SCIENTIFIC METHOD

This step-by-step process involved in learning about the sun is known as the **scientific method**. Scientists use this scientific method in their research work.

Let us study in detail the steps involved in the scientific method.

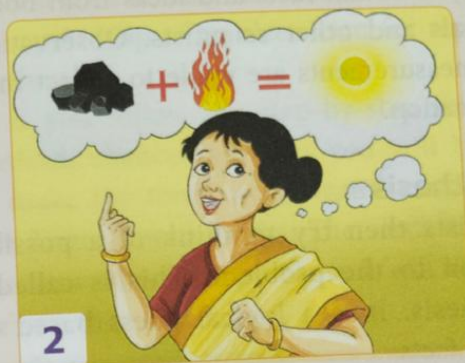
Problem

Scientists encounter something they cannot understand or explain. They first state the problem.

LONG AGO PEOPLE HAD NO IDEA WHAT THE SUN WAS AND HOW IT GOT ITS HEAT.



What is this huge bright ball? Why is it so hot? They wondered. The nature of the sun was a **problem** for them.



Some people thought that there was a huge amount of coal burning on the sun. This was a **hypothesis**. But this hypothesis could not explain why the sun had been burning for billions of years. So it was rejected.



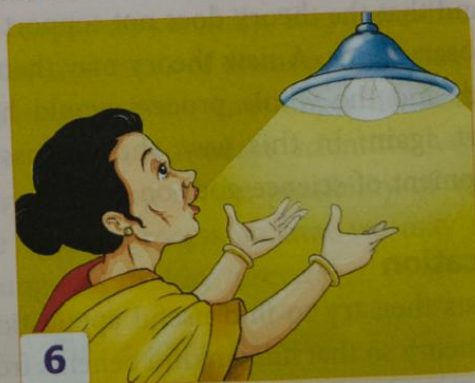
They made **observations** and **collected information**. They realized that the sun was one of the billions of stars in the universe.



Another hypothesis was made, based on more observations—the source of the sun's energy was hydrogen atoms which came together to form helium. **Experiments** were carried out to test the hypothesis.



This hypothesis was found to be correct. It now became a **theory**.



Scientists are trying to **apply** this theory in many ways. One of the possible **applications** is to generate electricity.

Observation and collecting information

Scientists obtain facts and ideas from books, journals and other scientists. Observations and measurements are made to collect more information.

Hypothesis

Scientists then try to think of a possible solution to the problem. This is called a hypothesis. It is a logical guess based on known facts.

Experiment

Scientists carry out further observations and experiments to test if the hypothesis is correct. In an experiment, observations and measurements are made under certain conditions. If they notice that the hypothesis is not correct, they may modify it or try to find a completely new hypothesis.

Theory

Once the hypothesis is proved to be correct, it becomes the conclusion or theory.

In course of time as better scientific tools and instruments become available, more and more observations may be made. It may then be found that the theory does not support the new observations. A new theory may then be needed, and the whole process would have to start again. In this way, the process of development of science goes on.

Application

Scientists then try to find useful applications of the theory so that humankind benefits from it. For example, the theory that if a magnet is moved near a coil of wire, an electric current

flows in the wire, has found applications in the form of **generators** that supply electricity to our houses.

Scientists rarely work alone. They mostly work together in teams. Often several teams around the world try to solve the same problem and consult one another. When several scientists think about how to solve a problem, a solution is easier to find. Similarly, if you work as a group with friends in trying to solve a problem, you will come up with a solution much faster.



THINK AND DISCUSS!

Reflective thinking

What is the purpose of using the 'scientific method'?

WHAT IS TECHNOLOGY?

When scientists make new discoveries, people work to make use of the discoveries to develop useful things. For example, rays called **X-rays**, which could pass through many materials, including muscles in the human body, were discovered in 1895. It was realized that X-rays could be used to take photographs of the bones inside the human body. Soon the **X-ray machine** was developed. While science deals with discovering new principles, **technology** deals with using these principles to make things that can do useful jobs. A scientific discovery has the potential of leading to the development of new technologies. People who develop new technologies are known as **technologists**.

THE SCIENTIFIC METHOD AND YOU

Scientists are people who follow the scientific method to solve problems or to discover

new things. You too can work like a scientist by using this method to learn about things around you. You only need an inquisitive mind and clear logical thinking.

For example, suppose your torch stops working. You encounter a problem—how to repair the torch. You do not know much about how a torch works, so you collect information by reading books or consulting an adult or an electrician. You find out that the torch contains cells that provide electric current, a bulb or LED that lights up when current is passed through it, a switch that turns the current on and off, and connecting wires inside the torch.

You next think of the possible reasons for the torch not working. You may think of any of these reasons:

- a. There are no cells in the torch.
- b. The cells have run out.
- c. The bulb or LED has fused.
- d. The switch is not working properly, or some wire inside is broken.

All these are your hypotheses. To check which hypothesis is correct, you will need to make observations and carry out certain experiments.

- ❖ Check if the cells are there in the torch. If not, your hypothesis **a** is correct. Put in cells.
- ❖ If the cells are present, take them out and put in new ones. If the torch now lights up, hypothesis **b** is correct.
- ❖ Check the bulb or LED. If it does not light up when connected to an electric cell, hypothesis **c** is correct. Change the bulb or LED.
- ❖ If none of these hypothesis **a**, **b** or **c** is correct, then there must be something wrong with the switch or the wires. You may, however,

realize that you do not have the expertise to repair or change the switch or wires. You then decide that it is better to get the expert help of an electrician for repairing the torch.

If you follow the step-by-step scientific method, chances are that you will be able to make the torch work again yourself and also learn how it works. The next time you encounter the same problem, it will be easier for you to solve.



ACTIVITY 1 (Observation and inference)

Think of a similar problem that you encountered and tried to solve. What steps did you take?

SCIENCE AND US

Today there are many scientific gadgets around us which help us in every activity. How many times do we stop to think of the years of research carried out by inquisitive minds that is behind the conveniences we take for granted today? Some of the main areas in which science has been especially beneficial to us are:

Health

Scientists have studied our body and life processes very carefully. They have discovered causes and cures of diseases. Today we expect to live a healthier and longer life because of this. In fact, nowadays scientists have even started talking of the possibility of stopping the ageing process so that we may remain young and healthy forever!



THINK AND DISCUSS!

Creative thinking

Suppose scientists were able to extend the lifespan of people to say 150 years. What advantages and disadvantages would this have?

Communication and transport

This is an area that has seen tremendous progress in the last few years, helped mainly by smaller and more powerful **computers**. We use computers to send letters through **e-mail**. We can get information on topics of our choice from libraries in Europe or the USA through the **Internet**. Cricket matches being played on the other side of the globe are shown to us on TV through **satellites**. Satellites also send data about the weather which is analyzed by **supercomputers** to predict the weather. Computers help us book our railway and air tickets. Today we can book our air tickets from our mobile phone even while sitting in a restaurant!

Travelling time has reduced considerably, due to aeroplanes that are controlled by computers and railway signalling systems that use computers.

Energy

With the possibility of coal and petroleum deposits disappearing in the near future, scientists are busy finding alternate sources of energy. Already solar energy, wind energy and tidal energy are being successfully used to generate electricity. Other sources of energy are being actively investigated. The Indian government has planned to make the desert area of Rajasthan one of the biggest electricity generating areas in the world, using solar energy.

There are innumerable other areas where science helps us in our daily life. You only need

to observe things around you, in your home, on the road, in your school, in a supermarket, to realise this.

THE ILL-EFFECTS OF SCIENCE

We have to pay a price for everything—we have certainly paid a price for scientific progress also. Long ago, people thought that the conveniences offered by science would help human beings to live a more comfortable and tension-free life. Though science has brought comforts, it has also increased tensions. Because of scientific progress, life has become so fast-paced that everyone is more busy and tense. No wonder, incidences of high blood pressure and heart attacks have increased. Scientific progress has caused an imbalance in nature. The natural resources in the world are getting more and more polluted. This could have disastrous consequences. The weapons of mass destruction created with the help of science may wipe out the human race from the earth.

IS SCIENCE GOOD OR BAD?



THINK AND DISCUSS!

Critical thinking

So, what do you think—is science good or bad?

The answer is—neither. Science is a tool in our hands. We may use it wisely to make the world a better place to live in, or we may use it foolishly to destroy what we have built up over centuries. The choice is ours!

NEW WORDS

HYPOTHESIS—a probable solution to a problem

THEORY—a correct hypothesis

TECHNOLOGY—using scientific theories to make things that can do useful jobs for us